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07.05.2024 (E)

SOMAIYA
VIDYAVIHAR UNIVERSITY

Maximum Marks: 100	Semester:– January 2024 – April 2024 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: UG (B.Tech)	Class: TY	Semester: VI (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering		Name of the department: Computers
Course Code: 116U01C601	Name of the Course: Digital Signal and Image Processing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	20
i)	Determine whether the following signal is periodic or not: $x(n) = sin\frac{\pi}{8}n^2$	5
ii)	Given is a 3*3 image, plot its bit planes. <table><tr><td>9</td><td>10</td><td>8</td></tr><tr><td>11</td><td>12</td><td>15</td></tr><tr><td>13</td><td>14</td><td>9</td></tr></table>	5
iii)	Explain in short different types of discrete time signals (any five).	5
iv)	Explain different mathematical operations on signals.	5
v)	Write a short note on digital negative.	5
vi)	Determine even and odd parts of the signal: $x(n) = \{\underset{\uparrow}{2},-2,6,-2\}$	5

Que. No.	Question	Max. Marks
Q2 A	Test the following systems for time invariance:	10
i)	$y(n) = x(n) + x(n-1)$	05
ii)	$y(n) = 2nx(n)$	05
	OR	
Q2 A	Construct the block diagram and signal flow graph of the discrete time system whose input-output relations are described by following difference equation $y(n) = 0.4y(n-1)+x(n)-3x(n-2)$	10
Q 2 B	Solve	10
i)	An 8 point sequence is given by x(n) = {2,1,2,1,1,2,1,2}. Compute 8-point DFT of x (n) by radix-2 DIT-FFT.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	20
i)	Explain the following spatial enhancement techniques with suitable example	10

	and state one application of each. a) Contrast stretching b) Log Transformation	
ii)	Compute the discrete cosine transform (DCT) matrix for N = 4.	10
iii)	Explain Low-pass Filtering in Frequency Domain	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Describe Canny Edge Detector in detail with an example.	10
ii)	Using Hough transform show that the following points are collinear. Also find the equation of the line for (x,y) plane are (1,2) ; (2,3) and (3,4).	10
iii)	Explain different Morphological operations with necessary equations.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / ~~Short question type~~) on any **four**	**20**
i)	Run Length Encoding.	5
ii)	JPEG Compression	5
iii)	Hoteling Transform	5
iv)	Vector Quantization	5
v)	Region Split and Merge based segmentation	5
vi)	Image Moments	5

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09.05.2024(E)

Maximum Marks: 100	Semester: January 2024–May 2024 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01 Programme: BTech in computer Emgineering	Class: _TY	Semester: VI(SVU 2020)	
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: COMP		
Course Code: 116U01E626	Name of the Course: Machine Lerning		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	How Gini index is used in Decision tree algorithm?	5
ii)	What is the principle of k-nearest neighbour?	5
iii)	What is the need of dimensionality reduction techniques? Describe different types of dimensionality reduction techniques.	5
iv)	What is the principle of k-fold cross validation technique?	5
v)	How confusion matrix is used to measure classifier accuracy?	5
vi)	What is underfitting and overfitting. How overfitting is avoided in decision tree algorithm?	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following.	**10**
i)	What is the difference between Markov model and hidden Markov model?	5
ii)	What are the different methods of data preprocessing?	5
	OR	
Q2 A	What is the principle of linear regression? Find the linear relationship between study time and marks obtained for the following dataset. Predict the marks of the student, studied for 800 minutes.	10

Sr no	Study time (min)	Marks obtained
1	350	520
2	1070	1600
3	630	1000
4	890	850
5	940	1350
6	500	690
7	550	700

Que. No.	Question	Max. Marks
Q 2 B	Solve any **One**	**10**
i)	What is Principle component analysis? Find out covariance matrix for the following dataset. Find out any one of the eigen vector. (solve upto one value of eigen vector calculation).	10

x	y
2.5	2.4
0.5	0-7
2.2	2.9
1.9	2.2

	3.1	3	
	2.3	2.7	
	2	1.6	
	1	1.1	
	1.5	1.6	
	1.1	0.9	
ii)	How Naïve Bays classifier works? Consider the following data and apply Naïve Bays classifier to find the different probabilities. For the given dataset. Find the probability of brand X with respect to Red Trouser.		10

	Cloth type	Brand	On sale?
Red	Trouser	Y	Yes
Red	Shirt	X	No
Red	Trouser	Y	Yes
Black	Trouser	Y	Yes
Black	Trouser	X	No
Black	Shirt	X	Yes
Black	Shirt	Y	Yes
Red	Trouser	Y	No
Black	Trouser	X	Yes
Red	shirt	X	Yes

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	How feed forward network and back propogation network works.?	10
ii)	How SVM works to identify hyperplane? How the margin is decided to separate hyperplanes?	10
iii)	What is the principle of bayesian belief network? What are the applications of bayesian belief network?	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Apply k-means clustering to the following data. Consider k = 2. C1 =(3,4) c2 = (6,4)	10
ii)	How HMM is used as the classifier after learning. Describe with suitable example.	10
iii)	How single linkage is applied in Agglomerative Algorithm (hierarchical method) Consider following samples. Apply clustering using single linkage technique.	10

Table for Q4 i):

Sample no	X	Y
1	2	6
2	3	4
3	3	8
4	4	2
5	6	2
6	6	4

Table for Q4 iii):

Sample No	X	Y
P1	0.40	0.53
P2	0.22	0.38
P3	0.35	0.32
P4	0.26	0.19

Que. No.	Question	Max. Marks
Q5	Solve any **four**	**20**
i)	How soft boundary is considered in SVM.	5
ii)	How KNN is different than K-means clustering?	5
iii)	How bottom up approach is used in hierarchical clustering?	5
iv)	How regression is different than classification? Justify with example.	5
v)	What are the elements of reinforcement learning?	5
vi)	What is the role of radial basis functions in separating nonlinear patterns?	5

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09.05.2024 (E)

Maximum Marks: 100	Semester: January 2024 –April 2024 Examination: ESE Examination		Duration:3 Hrs.
Programme code: 01 Programme: Computer Engineering	Class: TY B Tech		Semester: VI (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering		
Course Code: 116U01E637	Name of the Course: Microservices Foundations		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary			

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	Discuss the history of Microservices	5
ii)	Explain aspect of software application development with respect to development process and application architecture.	5
iii)	Explain how monolithic application requires more down time and inability to scale individual components?	5
iv)	List the components of Service Oriented Architecture.	5
v)	What are the different protocols are used in communication of microservices?	5
vi)	Explain how CI/CD helps in microservices development?	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Discuss key challenges for testing microservices	5
ii)	Explain the scenario - Testing between microservices internal to an application or residing within same application	5
	OR	
Q2 A	What are the types of microservices testing? Explain each one with an example.	10
Q 2 B	Solve any **One**	**10**
i)	What is use of API Gateway in microservices? How routing with an API Gateway is implemented?	10
ii)	Explain Three-Tier Spring Boot Application development with an example.	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain the domain driven design with the help of online retail example.	10
ii)	What is the use of data persistence pattern in microservices? Explain with the help of an example.	10
iii)	What are the benefits and challenges of domain driven design?	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Compare monolithic architecture and microservice architecture based of the following parameters. 1. Unit design 2. Communication within application 3. Technological flexibility 4. Deployment 5. Resiliency	10
ii)	What are the components of service oriented architecture? Explain each one.	10
iii)	When to use microservices? List the disadvantages of microservices.	10

Que. No.	Question	Max. Marks
Q5	Write notes on any **four**	**20**
i)	List any five design pattern for developing microservices	5
ii)	What is distributed architecture? Explain with the help of an example	5
iii)	Spring boot and spring cloud framework	5
iv)	Explain - *Build, Release, Run* - factor of application development.	5
v)	What are the use cases of microservices? Explain any one.	5
vi)	Building blocks of Domain Driven design	5

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09.05.2024(E)

Maximum Marks: 100	Semester: January 2024 –April 2024 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: B.Tech	Class:TY	Semester:VI(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code:116U01E629	Name of the Course: Cloud Computing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	What are the key characteristics of cloud computing?	5
ii)	Describe the Infrastructure as a Service (IaaS) model.	5
iii)	Differentiate Public cloud and Private cloud.	5
iv)	Discuss third party cloud Services.	5
v)	Explain Google Cloud Platform.	5
vi)	Describe NIST (National Institute of Standards and Technology) cloud definition.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Explain Cloud Reference model.	5
ii)	Write a note on Manjrasoft Aneka.	5
	OR	
Q2 A	Compare the compute, storage, and communication services of AWS and Google Cloud Platform?	10
Q 2 B	Solve any **One**	**10**
i)	Describe the role of virtualization in cloud computing?	10
ii)	Discuss some popular cloud-computing providers, and what distinguishes them from one another?	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	Explain the need of Virtualization in the context of Cloud Computing. Compare type-1 and type-2 Virtualization	10
ii)	Explain the concepts of Dockers, containers, and microservices in cloud computing?	10
iii)	Discuss the key features and benefits of Microsoft Hyper-V in the context of virtualization?	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	20
i)	Explain in brief about Amazon web services (AWS).	10
ii)	Illustrate the concept of scientific application using cloud for ECG analysis.	10
iii)	Discuss some challenges and considerations when deploying multiplayer online gaming applications in the cloud?	10

Que. No.	Question	Max. Marks
Q5	Write notes on any **four**	20
i)	Hadoop	5
ii)	SalesForce.com	5
iii)	Google AppEngine	5
iv)	ERP	5
v)	SpotCloud	5
vi)	DevOps	5

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09-05-2024 (E)

Semester: January 2024 –April 2024 Section B Maximum Marks: 50 Examination: ESE Examination Duration:2 Hrs.		
Programme code: _____ Programme: Computer Engineering	Class: TY	Semester: VI(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering	
Course Code: 116U01E629	Name of the Course: Cloud Computing	
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q 4	Solve any **Four**	20
i)	Discuss Amazon Web Services (AWS).	5
ii)	Explain Google Cloud platform (GCP).	5
iii)	Explain Google App Engine (GAE).	5
iv)	Explain the concept of Microsoft Hyper-V.	5
v)	Explain Cloud Reference Model.	5
vi)		5

Que. No.	Question	Max. Marks
Q 5	Write notes on any **four**	20
i)	ERP.	5
ii)	SpotCloud.	5
iii)	DevOps.	5
iv)	ECG using Cloud Computing.	5
v)	Multiplayer Online Gaming.	5
vi)	Protein Structure Prediction.	5

Que. No.	Question	Max. Marks
Q 6	Objective/Multiple Choice Questions	10
1	Which Google Cloud service is designed for real-time messaging and event ingestion? a) Google Cloud Pub/Sub b) Google Cloud Storage c) Google Cloud Datastore d) Google Cloud SQL	
2	What is Cloud Dataflow used for in Google Cloud Platform? a) Real-time data analysis and processing b) Virtual machine management c) Container orchestration d) File storage	
3	What is a hypervisor? a) A software that manages virtual machines and their resources b) A physical server c) A type of virtual machine	

	d) A network protocol	
4	What is virtualization? a) A technology that creates multiple virtual instances of physical hardware b) A technique used to simulate physical environments in software c) The process of converting physical servers into virtual machines d) All of the above	
5	Which of the following is a benefit of server virtualization? a) Increased hardware costs b) Decreased server utilization c) Improved resource utilization d) Limited scalability	
6	What is live migration in virtualization? a) Moving a virtual machine from one physical server to another without disrupting service b) Shutting down a virtual machine for maintenance c) Creating a new virtual machine from a template d) Allocating additional resources to a virtual machine	
7	What is Amazon S3? a) A relational database service b) A serverless computing platform c) An object storage service d) A content delivery network	
8	What is Amazon EC2? a) An object storage service b) A serverless computing service c) A virtual machine service d) A content delivery network	
9	What is Amazon RDS? a) A NoSQL database service b) A data warehousing service c) A managed relational database service d) A serverless computing service	
10	Which virtualization technology allows multiple virtual servers to share physical hardware resources while maintaining isolation? a) Network virtualization b) Storage virtualization c) Server virtualization d) Desktop virtualization	

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11/05/2024 (E)

Maximum Marks: 100	Semester: January 2024 – April 2024 Examination: ESE Examination	Duration: 3 Hrs.
Programme code: 01 Programme: B.Tech Computer Engineering	Class: T.Y.B.Tech	Semester: VI (SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer Engineering.	
Course Code: 116U01C602	Name of the Course: Information Security.	

Instructions: 1) Draw neat diagrams, wherever applicable. 2) All questions are compulsory. 3) Assume suitable data wherever necessary.

Que. No.	Question	Max. Marks
Q1	Solve *any Four* of the following:	20
i)	*"Security professionals balance the cost and effectiveness of controls with the likelihood and severity of harm."* Explain with supporting example – harm, likelihood and control in the above statement.	5
ii)	Interpret the outcome of the following: and justify the statement $D(E(M,K_U), K_U) = M = E(D(M,K_U),K_U)$ where M is the message, E is encryption function, D is the decryption function and K is the key for user U.	5
iii)	List three controls that could be applied to detect or prevent off-by-one errors.	5
iv)	Are computer-to-computer authentications subject to the weakness of replay attack? Why or Why not? Justify.	5
v)	List down the weaknesses of Wired Equivalent Privacy (WEP).	5
vi)	What are trade secrets? What are the two main requirements for information to be protectable as a trade secret?	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following:	10
i)	Write the strengths and weakness of DES.	5
ii)	Explain the Feistal structure of DES.	5
	OR	
Q2 A	A program is written to compute the sum of the integers from 1 to 10. A programmer, well trained in reusability and maintainability, writes the program so that it computes the sum of the numbers from k to n. However, a team of security specialists scrutinizes the code. The team certifies that this program properly sets k to 1 and n to 10; therefore, the program is certified as being properly restricted in that it always operates on precisely the range 1 to 10. List and explain the different ways that this program can be sabotaged so that during execution it computes a different sum, such as 3 to 20.	10
Q 2 B	Solve *any One*:	10
i)	Explain different compile-time and run-time defences for buffer overflow.	10
ii)	Explain the flaw and countermeasures for the following: (a) Undocumented Access Point. (b) Integer Overflow.	10

Que. No.	Question	Max. Marks
Q3	Solve *any Two*:	**20**
i)	If you forget your password for a website and you click (forgot my password), sometimes the company sends you a new password by email but sometimes it sends you your old password by email. (a) Compare these two cases in terms of vulnerability of the website owner. (b) Mention the countermeasures for overcoming these vulnerabilities.	10
ii)	What do you understand by clickjacking? Explain a technique by which a browser could detect and block clickjacking attacks.	10
iii)	What are email-based attacks? Explain any one such attack and the countermeasures for prevention of the same.	10

Que. No.	Question	Max. Marks
Q4	Solve *any Two*:	**20**
i)	List and explain any three threats to network communication.	10
ii)	With reference to network security, explain the following terms: (a) Port Scanning. (b) Any one tool for Port Scanning with features. (c) Outcome/Result of Port Scanning.	10
iii)	Write the different functionalities of a firewall. Why is firewall a good place to implement a VPN? Explain.	10

Que. No.	Question	Max. Marks
Q5	Write Short notes on *any four:*	**20**
i)	Digital Signatures.	5
ii)	Ransomwares.	5
iii)	Authentication mechanisms.	5
iv)	Link Encryption Vs End-to-End Encryption.	5
v)	Demilitarized Zones (DMZ).	5
vi)	Difference between Laws and Ethics.	5

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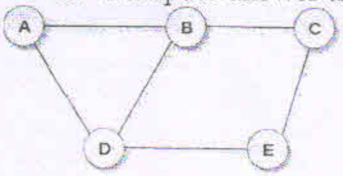
14-05-2024(E)

Maximum Marks: 100	Semester: January 2024 –April 2024 Examination: ESE Examination	Duration:3 Hrs.
Programme code: 01 Programme: Btech Computer Engineering	Class: TY	Semester: VI(SVU 2020)
Name of the Constituent College: K. J. Somaiya College of Engineering	Name of the department: Computer	
Course Code: 116U01C603 Name of the Course: Artificial Intelligence		
Instructions: 1)Draw neat diagrams 2) All questions are compulsory 3) Assume suitable data wherever necessary		

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	What are basic components of Intelligent systems?	5
ii)	Discuss and state the limitation of "**Thinking rationally approach of AI**"	5
iii)	Explain PEAS for automated driving scenario.	5
iv)	Discuss any two applications of AI.	5
v)	What is the advantage of Expert systems?	5
vi)	Give action description for wearing socks and shoes using ADL- Action Description Language.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	10
i)	When to use Bayesian Networks? Draw topology and conditional probability table for grass being wet in the event of : a. Sprinkler being on, or b. It is raining or c. Sprinkler being on and it is raining, both. Assume suitable data	5
ii)	Compare and contrast between A* and AO* search algorithm, give a suitable example to support your answer.	5
	OR	
Q2 A	The goal here is to assign each letter a digit from 0 to 9 so that the arithmetic works out correctly. The rules are that all occurrences of a letter must be assigned the same digit, and no digit can be assigned to more than one letter. Use Constraint Satisfaction Problem with backtracking (CSP) to: A. Solve the given instance of crypt-arithmetic problem. B. State constraints clearly C. State heuristic chosen to solve the problem. D. Show step by step solution using the chosen heuristic.	10

	TWO + TWO FOUR	
Q 2 B	Solve any **One**	**10**
i)	Discuss alpha-beta pruning as improvement over min-max algorithm. Discuss how the values of alpha and beta are chosen for a two player game with a suitable example.	10
ii)	Consider an instance of graph coloring problem, wherein any two adjacent vertices cannot have the same color. Assuming the number of given colors =3, A. Solve the same using genetic algorithms. A sample chromosome could be: (1,2,1,2,1) wherein 1 and 2 are color numbers. You can use random numbers to assign the colors and use number of attacking pairs as fitness function. B. Show contents of all steps clearly. C. Compute answer till one generation.  Undirected Graph	10

Que. No.	Question	Max. Marks
Q3	Solve any **Two**	**20**
i)	What do you mean by Intelligent agents? Explain simple reflex agent as a solution for predictive text suggestion in mobile keyboard.	10
ii)	1. Gita likes all kinds of food. 2. Mango and chapati are food. 3. Gita eats almond and is still alive. 4. Anything anyone eats and isn't killed by it is food. • **Goal:** Gita likes almond. Represent in First order logic and use forward chaining to prove the goal. State all inference rules used clearly.	10
iii)	Explain typical steps for Natural Language Processing.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	Describe Hill Climbing Search algorithm. What are the problems faced by Hill Climbing Search ? Explain the solutions too.	10
ii)	Describe steps in knowledge engineering process with respect to detection of Covid-19 infection when symptoms are given.	10

iii)	Design expert system for automated driving in mountains. Draw expert system component diagram and clearly state sample contents of them. State atleast 3 inference rules those might be part of the designed ES.	10

Que. No.	Question	Max. Marks
Q5	(Write notes / Short question type) on any **four**	**20**
i)	Explain the properties of task environment in Artificial Intelligence.	5
ii)	Consider a problem of arranging rainy season trip to Lonavla for a joint family of 10, ranging from ages 1 to 70. The trip would be planned for mid-June. A. Give atleast 2 plans of trip. B. What could be the uncertainties in the above plan.	5
iii)	Differentiate between planning and searching.	5
iv)	Differentiate between Uninformed search and Informed search strategy in AI.	5
v)	Explain the terms universal quantifier and existential quantifier.	5
vi)	State and discuss the things those AI cannot do in technical forum.	5